

OSI Model -- Guided Notes

Unit 1.1.1 | CompTIA Network+ N10-009 Obj. 1.1 | N10-008 1.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The OSI model divides network communication into _____ layers, from the _____ layer at the bottom to the _____ layer at the top.

Question 2

Layer 1 devices -- such as hubs, repeaters, and _____ -- deal only with raw _____ and have no understanding of addresses or protocols.

Question 3

A switch operates at Layer _____ and uses _____ addresses to forward frames to the correct port, rather than flooding traffic to every device like a hub.

Question 4

Explain the difference between a MAC address and an IP address. Which OSI layer is each one associated with?

Question 5

At Layer 4, the two main transport protocols are _____ and _____. The first provides reliable, ordered delivery; the second prioritizes _____ over reliability.

Question 6

Port numbers exist at Layer _____. They allow a single device with one IP address to run multiple _____ simultaneously (for example, a web server and an SSH daemon on the same machine).

Question 7

When a browser negotiates a TLS connection to load an HTTPS website, which OSI layer handles that encryption setup? _____

Question 8 -- Real World Application

REAL WORLD: A user reports that they can ping 8.8.8.8 successfully but cannot load any websites in their browser. Based on the OSI model, which layer(s) would you investigate first, and what specific service is most likely the cause? Explain your reasoning in two to three sentences.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-1-1-osi-model/1-1-1-osi-model-answer-key.pdf
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